

EE236: Electronic Devices Lab

Lab 10: C-V Characteristics of MOS Capacitor

Abhineet Agarwal, 22b1219

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1 Aim of the experiment

- Create a summer and amplifier circuit and test its working.
- Find the variation of CDUT with VDC and calculate the MOSCAP parameters- Oxide capacitance, Oxide thickness, Doping density, Flat band voltage, Flat band capacitance, Debye length, Debye capacitance.

2 Part-1: Summer Circuit

2.1 Circuit

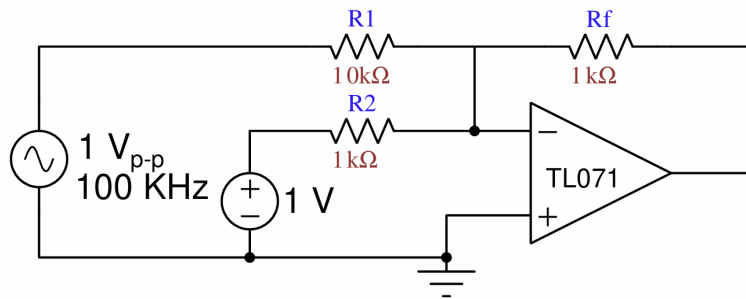


Figure 1: Summer Circuit

2.2 Experimental Results

The circuit returns the weighted sum of inputs correctly.

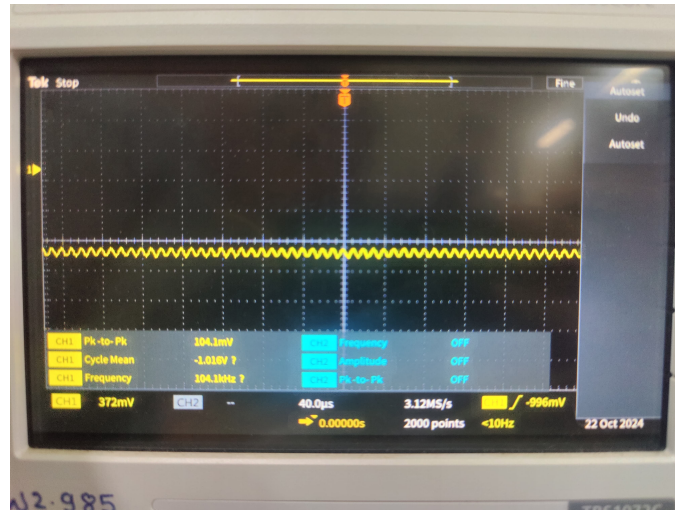


Figure 2: Summer circuit output

3 Part-2: Amplifier Circuit

3.1 Circuit

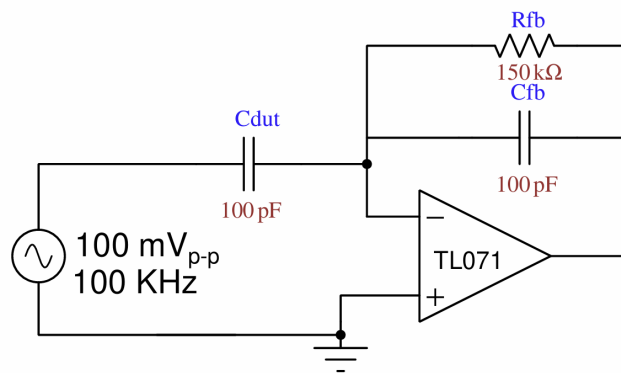


Figure 3: Amplifier Circuit

3.2 Experimental Results

The circuit works well at higher frequencies but starts to give shaky outputs when the frequency decreases beyond 5kHz. It acts as a high pass filter.

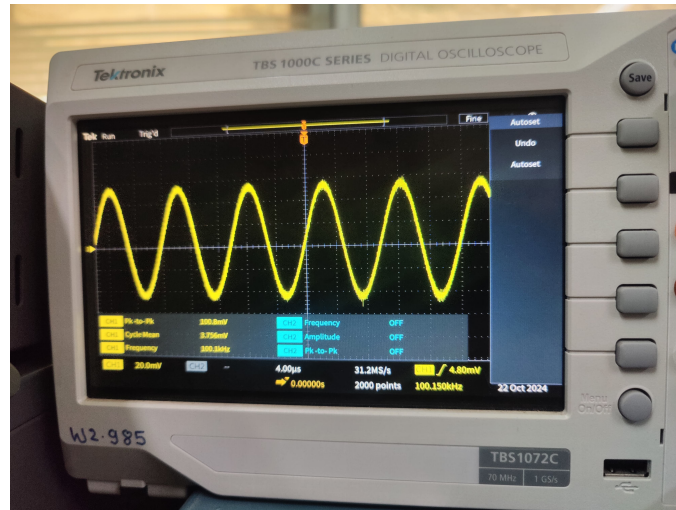


Figure 4: Output at 100kHz

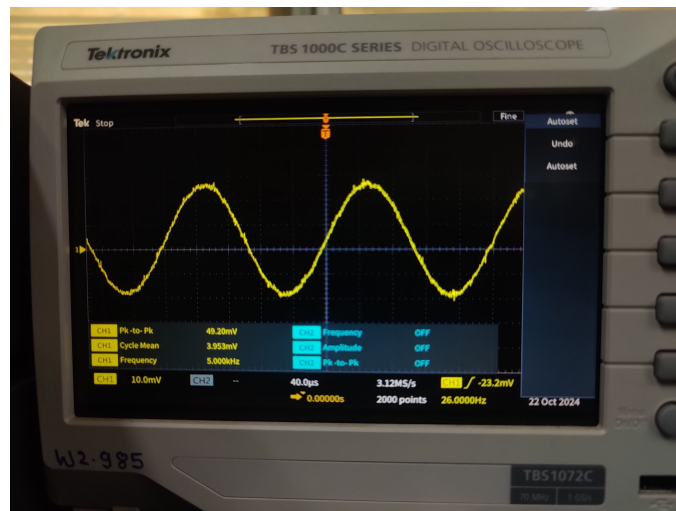


Figure 5: Output at 5kHz

4 Part-3: Measurement of MOSCAP C-V characteristics

4.1 Circuit

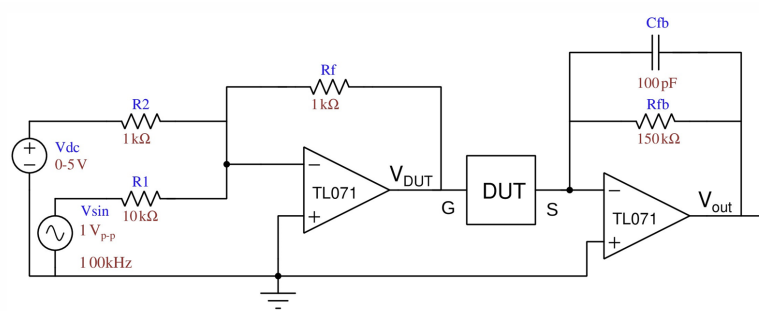


Figure 6: Circuit for CV characteristics measurement

4.2 Readings

Vdc	Vdut(p-p)	Vout	AC Gain	Cdut
-5	200	748	3.74	448.8
-4.5	200	748	3.74	448.8
-4	200	752	3.76	451.2
-3.5	200	752	3.76	451.2
-3	200	752	3.76	451.2
-2.5	200	752	3.76	451.2
-2	200	752	3.76	451.2
-1.5	200	752	3.76	451.2
-1	200	760	3.8	456
-0.5	200	760	3.8	456
0	200	760	3.8	456
0.5	196	776	3.959	475.102
1	196	776	3.959	475.102
1.5	208	792	3.808	456.923
2	220	860	3.909	469.091
2.5	200	1032	5.16	619.2
3	200	1460	7.3	876
3.5	240	1620	6.75	810
4	240	1650	6.875	825
4.5	240	1670	6.958	835
5	240	1680	7	840

Table 1: Data for circular NCAP with 2mm diameter

Vdc	Vdut (mV)	Vout (mV)	AC Gain	Cdut(pF)
-5	200	1680	8.4	1008
-4.5	200	1680	8.4	1008
-4	200	1680	8.4	1008
-3.5	200	1680	8.4	1008
-3	200	1660	8.3	996
-2.5	200	1660	8.3	996
-2	200	1660	8.3	996
-1.5	200	1660	8.3	996
-1	200	1660	8.3	996
-0.5	200	1640	8.2	984
0	200	1620	8.1	972
0.5	200	1260	6.3	756
1	200	740	3.7	444
1.5	200	732	3.66	408.2
2	200	560	2.8	336
2.5	200	516	2.58	309.6
3	200	484	2.42	290.4
3.5	200	472	2.36	283.2
4	200	472	2.36	283.2
4.5	200	468	2.34	280.8
5	200	468	2.34	280.8

Table 2: Data for circular PCAP with 2mm diameter

4.3 C_{DUT} vs V_{DC} curves

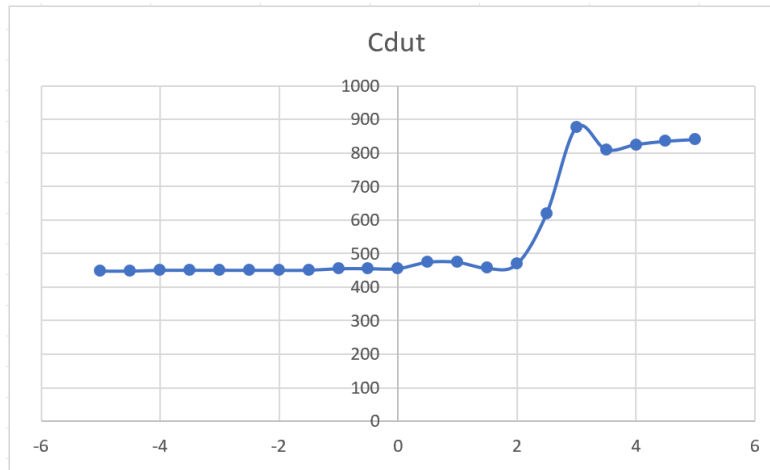


Figure 7: NCAP

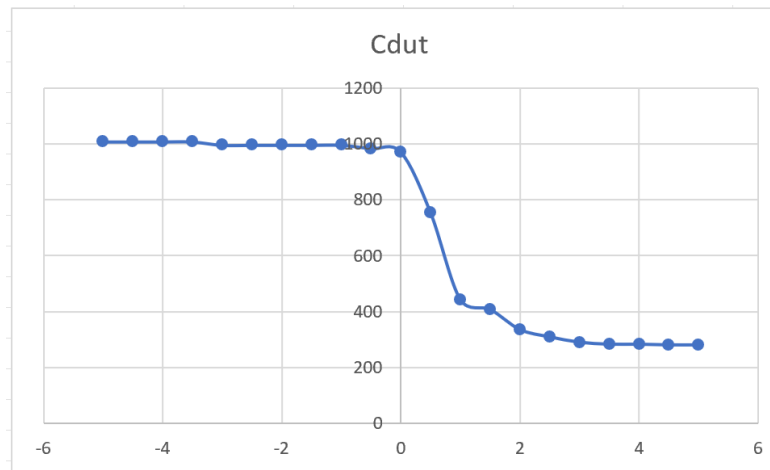


Figure 8: PCAP

5 Calculation of MOSCAP Parameters (for NCAP)

5.1 Standard Parameters

- Intrinsic carrier concentration of Silicon: $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$
- Silicon dioxide permittivity: $\varepsilon_{ox} = 3.45 \times 10^{-13} \text{ F cm}^{-1}$
- Silicon permittivity: $\varepsilon_{si} = 1.04 \times 10^{-12} \text{ F cm}^{-1}$
- Thermal voltage: $\frac{kT}{q} = 0.026 \text{ V}$ at room temperature
- Device area: $A = \pi r^2 = \pi(1 \text{ mm})^2 = 3.14159 \times 10^{-2} \text{ cm}^2$

5.2 Oxide Capacitance (C_{ox})

From the C-V measurements, the maximum capacitance is the oxide capacitance:

$$C_{ox} = 840 \text{ fF}$$

5.3 Oxide Thickness (t_{ox})

Using the equation:

$$t_{ox} = \frac{A\varepsilon_{ox}}{C_{ox}}$$

Substituting values:

$$t_{ox} = \frac{3.14159 \times 10^{-2} \text{ cm}^2 \times 3.45 \times 10^{-13} \text{ F cm}^{-1}}{840 \times 10^{-15} \text{ F}} \approx 12.9 \text{ nm}$$

5.4 Doping Density (N_A)

5.4.1 Depletion Width (t_{dep})

First, we use the depletion width equation:

$$t_{dep} = 2\sqrt{\frac{\varepsilon_{si}kT}{qN_A} \ln\left(\frac{N_A}{n_i}\right)}$$

5.4.2 Semiconductor Capacitance (C_s)

The semiconductor capacitance in strong inversion is related to the depletion width:

$$C_s = \frac{A\varepsilon_{si}}{t_{dep}}$$

5.4.3 Minimum Capacitance (C_{min})

From the C-V data, we observe:

$$C_{min} = 448.8 \text{ fF}$$

This is related to C_{ox} and C_s by:

$$C_{min} = \frac{C_{ox}C_s}{C_{ox} + C_s}$$

5.4.4 Solving for N_A

Using these relationships:

1. First, we can find C_s from C_{min} and C_{ox} :

$$C_s = \frac{C_{ox}C_{min}}{C_{ox} - C_{min}} = \frac{840 \times 448.8}{840 - 448.8} \approx 960 \text{ fF}$$

2. Then, we can find t_{dep} :

$$t_{dep} = \frac{A\varepsilon_{si}}{C_s} = \frac{3.14159 \times 10^{-2} \text{ cm}^2 \times 1.04 \times 10^{-12} \text{ F cm}^{-1}}{960 \times 10^{-15} \text{ F}} \approx 34.1 \text{ nm}$$

3. Finally, using the depletion width equation and solving iteratively:

$$34.1 \text{ nm} = 2\sqrt{\frac{1.04 \times 10^{-12} \text{ F cm}^{-1} \times 0.026 \text{ V}}{q \times N_A}} \ln\left(\frac{N_A}{1.5 \times 10^{10} \text{ cm}^{-3}}\right)$$

This yields:

$$N_A \approx 2.3 \times 10^{16} \text{ cm}^{-3}$$

5.5 Debye Length (L_{debye})

Using the equation:

$$L_{debye} = \sqrt{\frac{\epsilon_{si} kT}{qN_A}}$$

Substituting values:

$$L_{debye} \approx 28.7 \text{ nm}$$

5.6 Debye Capacitance (C_{debye})

Using the equation:

$$C_{debye} = \frac{A\epsilon_{si}}{L_{debye}}$$

Substituting values:

$$C_{debye} \approx 1130 \text{ fF}$$

5.7 Flat Band Capacitance (C_{fb})

Using the equation:

$$C_{fb} = \frac{C_{ox} C_{debye}}{C_{ox} + C_{debye}}$$

Substituting values:

$$C_{fb} = \frac{840 \times 1130}{840 + 1130} \approx 480 \text{ fF}$$

5.8 Flat Band Voltage (V_{fb})

From the C-V data, interpolating where capacitance equals C_{fb} :

$$V_{fb} \approx 1.0 \text{ V}$$

6 Calculation of parameters for PCAP

6.1 Oxide Capacitance (C_{ox})

From the C-V measurements, the maximum capacitance in strong accumulation:

$$C_{ox} = 1008 \text{ pF} = 1.008 \times 10^{-9} \text{ F}$$

6.2 Oxide Thickness (t_{ox})

Using the equation:

$$t_{ox} = \frac{A\epsilon_{ox}}{C_{ox}}$$

Substituting values:

$$t_{ox} = \frac{3.14159 \times 10^{-2} \text{ cm}^2 \times 3.45 \times 10^{-13} \text{ F cm}^{-1}}{1.008 \times 10^{-9} \text{ F}} \approx 0.0107 \text{ nm}$$

6.3 Doping Density (N_A)

6.3.1 Semiconductor Capacitance (C_s)

Using minimum capacitance from C-V data:

$$C_{min} = 280.8 \text{ pF} = 2.808 \times 10^{-10} \text{ F}$$

Calculate C_s :

$$C_s = \frac{C_{ox}C_{min}}{C_{ox} - C_{min}} = \frac{1008 \times 280.8}{1008 - 280.8} \approx 385 \text{ pF}$$

Calculate t_{dep} :

$$t_{dep} = \frac{A\epsilon_{si}}{C_s} = \frac{3.14159 \times 10^{-2} \text{ cm}^2 \times 1.04 \times 10^{-12} \text{ F cm}^{-1}}{3.85 \times 10^{-10} \text{ F}} \approx 0.085 \text{ nm}$$

6.4 Debye Length (L_{debye})

$$L_{debye} \approx 0.063 \text{ nm}$$

6.5 Debye Capacitance (C_{debye})

$$C_{debye} \approx 515 \text{ pF}$$

6.6 Flat Band Capacitance (C_{fb})

$$C_{fb} = \frac{C_{ox}C_{debye}}{C_{ox} + C_{debye}} \approx 340 \text{ pF}$$

6.7 Flat Band Voltage (V_{fb})

From the C-V data, interpolating where capacitance equals C_{fb} :

$$V_{fb} \approx 2.0 \text{ V}$$